# DATABASE SECURITY SETUP

## (Roles, Network Policy, and Data Masking)

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## 1. CREATE AND MANAGE USERS AND ROLES

### Step 1: Create Roles

```
CREATE ROLE admin;
CREATE ROLE analyst;
CREATE ROLE viewer;
```

### Step 2: Assign Privileges to Roles

```
GRANT ALL PRIVILEGES ON DATABASE sales_db TO ROLE admin;
GRANT SELECT ON ALL TABLES IN SCHEMA sales_db.public TO ROLE analyst;
GRANT USAGE ON WAREHOUSE compute_wh TO ROLE viewer;
```

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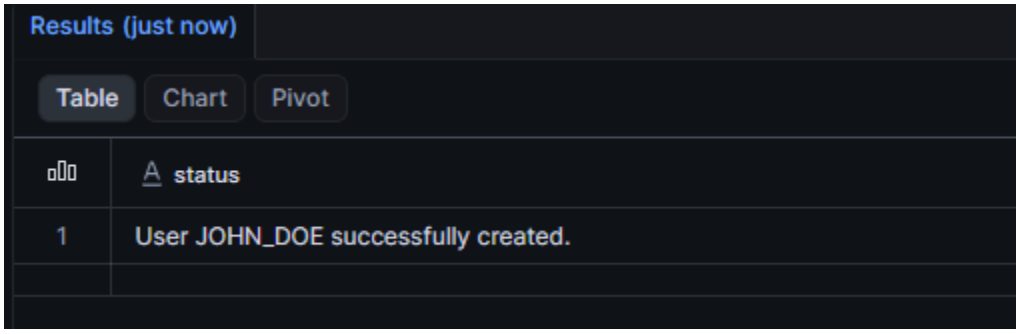
### Step 3: Create Users

```
CREATE USER john_doe
PASSWORD = 'Password123!'
DEFAULT_ROLE = analyst
MUST_CHANGE_PASSWORD = TRUE;
```

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### Step 4: Assign Roles to Users

```
GRANT ROLE admin TO USER admin_user;
```

```
GRANT ROLE analyst TO USER john_doe;
GRANT ROLE viewer TO USER jane_doe;
```

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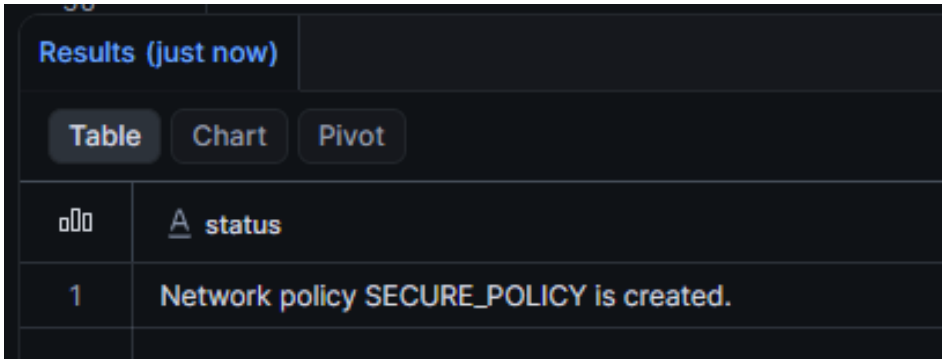
# 2. CONFIGURE NETWORK POLICIES

### Step 1: Create a Network Policy

```
CREATE NETWORK POLICY secure_policy
ALLOWED_IP_LIST = ('192.168.1.0/24', '203.0.113.0/32')
BLOCKED_IP_LIST = ('10.0.0.0/8');
```

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### Step 2: Apply Network Policy

```
ALTER ACCOUNT SET NETWORK_POLICY = secure_policy;
```

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# 3. IMPLEMENT DATA MASKING ON SENSITIVE DATA

### Step 1: Create a Table with Sensitive Data

```
CREATE TABLE employees (
  id INT,
  name STRING,
  ssn STRING
);

INSERT INTO employees VALUES (1, 'Alice', '123-45-6789');
```

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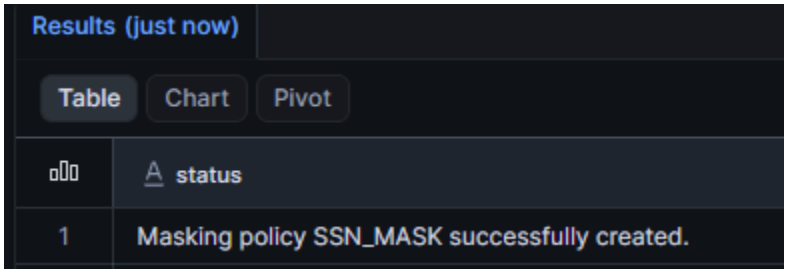
## Step 2: Define a Masking Policy

```
CREATE MASKING POLICY ssn_mask AS (val STRING) RETURNS STRING ->
CASE
  WHEN CURRENT_ROLE() = 'admin' THEN val
  ELSE 'XXX-XX-XXXX'
END;
```

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## Step 3: Apply the Masking Policy

```
ALTER TABLE employees
MODIFY COLUMN ssn
SET MASKING POLICY ssn_mask;
```

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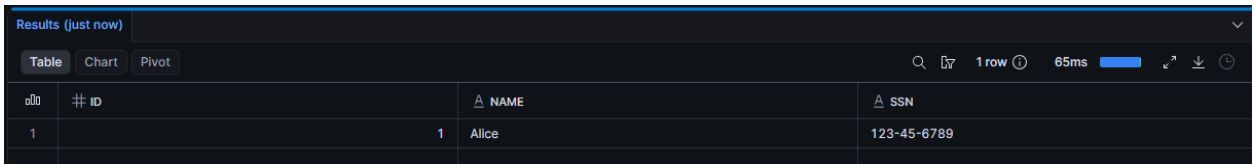
## Step 4: Test the Masking Policy

```
-- Admin role (unmasked data)

SET ROLE admin;
SELECT * FROM employees;
```




```
-- Analyst role (masked data)

SET ROLE analyst;
SELECT * FROM employees;
```